

Architectural Particular Specification
VINYL FLOORING
SECTION 8.7

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SECTION 8.7

VINYL FLOORING

PART 1 GENERAL

1.01 SUMMARY OF WORK

- A. The Work of this Section includes, but is not limited to, all the materials and labour for the furnishing and installation of the vinyl flooring as indicated on plans. It shall include the preparation of the base of the surfaces and incorporation of self-levelling compound necessary for the application.
- B. The vinyl sheet flooring system shall be a proprietary product and supplied by the same manufacturer for consistency and installed by authorized agents of the manufacturer.
- C. The Contractor shall prepare the detailed arrangement of the installation for the Architect's approval.

1.02 REFERENCES

- A. ASTM 1303-95 : Dimensional Stability
- B. ASTM F970-87 : Residual Indentation
- C. ASTM F137 : Flexibility
- D. ASTM E648-97 : Flame Spread
- E. ASTM D3884 : Abrasion Resistance
- F. ASTM C1028-89 : Coefficient of Friction

1.03 QUALITY ASSURANCE

- A. The manufacturer of the floor covering must be in possession of a valid quality system certificate showing compliance with ISO9001 quality system.
- B. The manufacturer of the floor covering must be in possession of a valid environmental certificate showing compliance with ISO14001 environmental management system.

1.04 PERFORMANCE REQUIREMENT

- A. The flooring system shall include all the works incorporating all necessary components, the adhesive and the preparation of the screed including the application of the leveling compound, necessary for the works.
- B. The Contractor shall make provision in the detailed arrangement for all additional customized components and elements to suit the actual requirements on site.
- C. The Contractor shall include for all finishing of the materials at junctions to fittings, including cupboards, plinths, shutters, joinery, tactile tiles, etc.
- D. Prior to installation, the Contractor shall check all actual dimensions on site based on final layout plans and other working details.

- E. The Contractor shall also be responsible for:
1. Shop drawings
 2. Schedule and monitoring of the works
 3. Samples and mock-up by request
 4. Co-ordination with works of the trade
 5. Protection
 6. Hoisting and temporary site work
- F. Installation of all works shall be in strict accordance with the manufacturer's specification and recommendation.
- G. The finished works shall be smooth, clean and free from damage, oil, grease, solvent and defects.
- H. Unless otherwise specified, all materials shall be of the best quality of the types specified.
- I. The Contractor shall ensure the complete sealing of all joints to prevent moisture ingress and the design of the system shall fully demonstrate and achieve this requirement.
- J. The Contractor shall be deemed to have fully allowed for selection of colours of flooring from the manufacturer's standard colour range as required by the Architect, forming patterns within a single area of flooring, or having different colours of flooring in different areas. All welded joints shall be of matching colours by multi-colour welding rod as approved by the Architect.
- K. The Contractor shall provide a high quality substrate for receiving the vinyl sheet flooring. The substrate shall be properly prepared strictly in accordance with the manufacturer's stipulations including the application of the self-levelling compound to all areas where the flooring is required.
- L. All joints shall be grooved and hot welded fully by welding rod in accordance with the manufacturer's instructions.
- M. The manufacturer of the entrance floor mat must be in possession of a valid quality system certificate showing compliance with ISO9001:2008 quality system.
- N. The manufacturer of the entrance floor mat must be in possession of a valid environmental management system certificate showing their operation is in compliance with ISO14001:2004 system.
- O. The manufacturer of the entrance floor mat must be in possession of a valid occupational health and safety management system certificate, showing their operation is in compliance with OHSAS 18001:2007 system.

1.05 ENVIRONMENTAL REQUIREMENTS

(NOT USED)

1.06 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITIONS

(NOT USED)

1.07 SUBMITTAL

A. The Contractor shall submit the followings:

1. The Contractor shall submit full vinyl sheet samples to the Architect for approval and colour and pattern selection.
2. The Contractor shall submit performance data of all materials and articles to be used in the work for the approval of the Architect.
3. The order for bulk supplies of materials and articles shall not be confirmed until formal approval is obtained from the Architect.
4. Irrespective of the materials and articles offered in the tender, all materials and articles for use of the works shall be in full compliance with the Specification unless otherwise agreed or approved by the Architect.

B. Shop drawings:

1. The Contractor's detailed arrangement for the System shall satisfy the performance requirements and include, inter alia, the following:-
2. Floor plans showing the configuration and layout of the flooring, including in particular the joints and welding positions where applicable and indicate penetration;
3. Assembly and all construction details;
4. Junction details with flooring itself and to the fittings;
5. Trimming details around openings and connection of floor drains and gratings penetrations.;
6. Integral cove skirting arrangements, including cutting and welding with welding rod arrangement at internal and external corners;
7. Construction details at changes in level;
8. Work of other installations;
9. Work by other trades.
10. Identify each material and product and the compatibility with the substrate.
11. Mounting detail with the appropriate adhesive for specific substrates.

C. Product Data

1. Submit manufacturer's technical data for materials, fabrication, finishing, fastenings, hardware and installation details.

D. Submit as-built record drawing and operation & maintenance manual

1.08 MOCK-UPS

This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.09 DELIVERY, STORAGE AND HANDLING

(NOT USED)

1.10 TESTING

(NOT USED)

1.11 WARRANTY

- A. This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.
- B. Minimum ten (10) years warranty against material defects and excessive wear and tear for normal use for vinyl floor.

1.12 SPARE PARTS

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PART 2 PRODUCTS

2.01 APPROVED MANUFACTURES/ SUPPLIERS

- A. Subject to compliance with the Project Contract Documents, provide vinyl sheet from the following approved manufacturers/ suppliers, or equivalent approved by the Architect:

1. The Jardine Engineering Corporation Limited
5/F., Tower A, Manulife Financial Centre,
223-231 Wai Yip Street, Kwun Tong,
Kowloon

Tel: 852 6293 5016 Fax: 852 2887 9000

2.02 MATERIALS

- A. The 2.0mm thick vinyl floor sheeting must be available in 2 meter width and 20 meter long according to ISO 24346- EN428 and ISO24340 – EN429 to minimize the number of joints.
- B. The rolls should be in 2m long according to ISO 24341 – EN426
- C. The total weight should be 3000g/m2 or less.
- D. The flooring shall meet ISO 10581 - EN649, and be classified as Class 34-43 according to ISO 10874 - EN685, i.e. suitable for very heavy commercial / institutional areas.
- E. The product must have polyurethane surface treatment for reduce the cost of maintenance and no polish required for the entire lifetime after installation.
- F. Regarding the Residual Indentation, the product should conforms to the requirements of ISO 24343-1 - EN433 with a testing result of $\leq 0.10\text{mm}$
- G. In respect of reaction to fire, the flooring shall have been fully tested to EN 13501-1 and certified as having Class Bfl-S1, meet $\geq 8\text{kW/m}^2$ according to EN ISO 9239-1 and pass the test of ENISO11925-2.
- H. Regarding the Dimension stability, the vinyl flooring should conforms to the requirements of ISO 23999 - EN434 with a testing result of $\leq 0.4\%$ for rolls and $\leq 0.25\%$ for tiles
- I. Castor resistance in accordance with ISO4918 - EN 425: suitable for office chairs with castors.
- J. The vinyl flooring shall provide anti-static properties EN 1815, $\leq 2\text{kV}$
- K. The vinyl flooring shall conform to the color fastness test of EN 20105-B02, ≥ 6 .
- L. The chemical resistance shall be good resistance according to ISO 26987 – EN423.
- M. The fungi and bacterial resistance shall be “Does not favor growth” according to ISO846: Part C.
- N. The clean room test shall be Class A according to ASTM F24-04 and ASTM F51-00.
- O. Self-leveling compound:
1. Full surface leveling of the subfloor self-levelling compound shall be applied. Such proprietary leveling compound shall be applied to achieve a smooth flat surface to receive vinyl flooring. The levelling compound shall be subject to

the Architect's approval, and shall be of a material that is compatible with the vinyl flooring, forming the whole of the flooring system. The application of the leveling compound shall be strictly in accordance with the manufacturer's stipulations.

2. Self-leveling floor compound shall be applied on top of the Water Based Epoxy Waterproofing Membrane, to the primed surfaces or directly to the prepared screed. The self-levelling floor compound shall comply with the properties as described below.
3. All screeds to receive vinyl sheet floor finish shall receive 5mm thick proprietary Self-Levelling Compound Surface. The self-levelling compound shall comply with the following performance specification:
4. The proprietary self- leveling compound shall be a machine or hand applied self-levelling compound designed as a smooth underlayment screed for using on concrete floor to receive vinyl sheets, carpeting and other floor finishes.

Product	Self-Leveling ST-SL (U)
Color	Grey
Comp. Strength (28D)	≥30 Mpa
Flexural Strength (28D)	≥ 7 Mpa
Tensile Adhesion Strength	≥1.5 Mpa
to Concrete Substrate (28D)	≥ 140 mm
Coat thickness in single Operation	5-25 mm
Water Demand (per 25kg bag)	~5.0-5.5 L
Coverage for Reference	~1.7 kg/m2/mm
Curing Time for foot traffic	~2-4 hours
Curing Time for covering	18-24hours

5. The self-levelling compound shall have the following characteristics:
 - a. Be supplied as dry powder designed for application at thickness of 3-5mm in one operation. The application thickness shall be not less than 0.5mm
 - b. Have fast setting time and shall be available for foot traffic after 2 – 3 hours at 25°C
 - c. Have single component with fixed mixing proportion for sure the quality of work.
 - d. Protein free and VOC content: < 5g / L, that is environmental friendly
6. The self-levelling compound should be able to allow foot traffic in 2 - 3 hours in 25°C. Direct stick impervious floor coverings such as vinyl can generally be applied after 48 hours.
7. The self-levelling compound shall be applied onto a clean sound and well-prepared substrate, Concrete floors must be solid, thoroughly clean and free of oil, wax, grease, asphalt, latex compounds, curing and sealing compounds and any other surface contaminant. Mechanically clean the floor to provide a roughened, clean sound solid and open porous matrix of the concrete, by using recommended preparation methods such as shot blasting, scarifying, diamond grinding/shaving or other suitable method. Acid etching is not an acceptable means of cleaning the sub-floor. Do not use solvents or

sweeping compounds to clean the sub-floor. Sub-Floors must be dry (A.S. 1884 1985) and properly primed for a successful installation. Sub-floor temperatures must be a minimum of 10°C and increasing for the installation.

8. A dry sub-floor means the relative humidity < 70% or moisture content < 5.5 %.

2.03 FABRICATION

(NOT USED)

2.04 WORKMANSHIP

(NOT USED)

PART 3 EXECUTION

3.01 EXAMINATION / INSPECTION

(NOT USED)

3.02 PREPARATION

- A. The subfloor must be properly prepared in strict compliance with the manufacturer's stipulations to present a fairly porous, dry, smooth surface that is free of cracks, abscesses and separation joints. A "Preliminary Moisture Tests" should be verified by an independent qualified testing authority.
- B. Prepare the subfloor with a suitable primer. At ground level floors and where floors are below ground level or about earth, or in areas where rising moisture may be expected, apply Water Based Waterproofing Membrane, strictly in accordance with the manufacturer's recommendations.
- C. Apply the flooring in accordance with the manufacture's recommendations. Ensure that all sheets are properly set out and that all cutting is done at the perimeter. Apply welding rods as necessary and carry out welding with multi-colour welding rod in full compliance with the manufacturer's recommendations.
- D. The contractor shall verify all as-built dimensions and conditions on site before commencing the works. The highest workmanship is required and all works shall be performed in an efficient and expedient manner by skilled workman experienced with the type of installation.
- E. Flooring joints shall be constructed square and true to plane, both in themselves and in relationship with each other where installed in a continuous plane. All components and materials shall be carefully handled at work and on site, both during delivery or handling. Any damaged or defective materials will be rejected.
- F. The sub floor must be absolutely hard, level, dry, smooth and structurally sound.
- G. The sub floor must be free of cracks and other irregularities and must not be contaminated with paint, plaster, oil, grease or any other substance which could affect adhesion.
- H. The sub floor must be finished with self leveling screeding.
- I. Cracks must be filled with an appropriate material.

3.03 INSTALLATION

A. Vinyl Flooring

1. Apply the flooring in accordance with the manufacture's recommendations. Ensure that all sheets are properly set out and that all cutting is done at the perimeter. Apply welding rods as necessary and carry out welding with multi-colour welding rod in full compliance with the manufacturer's recommendations.
2. Flooring joints shall be constructed square and true to plane, both in themselves and in relationship with each other where installed in a continuous plane. All components and materials shall be carefully handled at work and on site, both during delivery or handling. Any damaged or defective materials will be rejected.
3. Install all work square, plumb, straight, to line accurately fitted and located, with flush tight hairline joints if practicable (except as indicated otherwise where joints are to be welded).
4. The vinyl floor adhesive and smoothing compound used shall be fully to the vinyl manufacturer's recommendations to ensure full product compatibility.
5. Products must be fully conditioned with the environment in which they are to be installed as outlined by vinyl flooring manufacturer.
6. All joints in the vinyl sheet flooring system must be grooved and then hot welded by welding rod.
7. Installation must be carried out in accordance with BS 8203 and the instructions of the vinyl flooring manufacturer.
8. The floor for installation must be smooth, clean, dry, free of oil / grease and structurally sound. If direct-to-earth installation is required, an effective waterproof membrane must be incorporated. Moisture test shall be carried out to ensure that the subfloor has dried out to a level consistent with the application of vinyl flooring.
9. It is the responsibility of the Contractor that shop fabricated items shall properly fit the site conditions.
10. In cases where the shop fabricated items do not fit the site conditions, the items shall be returned for correction at the Contractor's own expense.

3.04 CLEANING

(NOT USED)

3.05 PROTECTION

(NOT USED)

END OF SECTION